

Table 11: Behavioral evaluation for strict subjects with $\alpha = 0.5$.

Concept	Regime	Leak (%)	Refuse (%)	Other (%)	n
book	hint	1.0	33.0	66.0	303
	refusal	0.0	92.5	7.5	306
	sametext	0.0	0.0	100.0	300
	think	0.6	57.3	42.1	309
	offtopic	0.3	0.3	99.3	300
flag	hint	0.0	34.7	65.3	303
	refusal	0.0	94.1	5.9	306
	sametext	0.0	0.0	100.0	300
	think	0.0	66.0	34.0	309
	offtopic	0.0	0.7	99.3	300
leaf	hint	0.7	34.3	65.0	303
	refusal	0.0	92.5	7.5	306
	sametext	0.0	0.0	100.0	300
	think	0.0	56.6	43.4	309
	offtopic	0.0	0.7	99.3	300
moon	hint	0.0	42.9	57.1	303
	refusal	0.0	95.8	4.2	306
	sametext	1.0	0.0	99.0	300
	think	0.0	63.4	36.6	309
	offtopic	0.0	1.0	99.0	300
wave	hint	0.3	37.0	62.7	303
	refusal	0.0	94.1	5.9	306
	sametext	0.0	0.0	100.0	300
	think	0.3	69.3	30.4	309
	offtopic	0.3	0.3	99.3	300

gregating across subjects and regimes yields the exact-recovery bars in Figure 5.

J.5 Detailed target-probability results

Tables 25–28 report per-cell $\overline{P}(c^*)$ for the Base AO, Cross FT-AOs, and Own FT-AO. Cross entries report mean [min, max] over the non-matching FT-AOs available for that (protocol, α) cell. A subscript indicates the number of cross AOs when fewer than four were available. Aggregating across subjects and regimes yields the target-probability bars in Figure 5.

J.6 Detailed median-rank results

Tables 29–32 report per-cell median target rank for the Base AO, Cross FT-AOs, and Own FT-AO. Cross entries report mean [min, max] over the non-matching FT-AOs available for that (protocol, α) cell. A subscript indicates the number of cross AOs when fewer than four were available. Aggregating across subjects and regimes yields the median-rank bars in Figure 5. Lower rank indicates greater target accessibility.

Table 12: Behavioral evaluation for strict subjects with $\alpha = 1.0$.

Concept	Regime	Leak (%)	Refuse (%)	Other (%)	n
book	hint	1.3	15.5	83.2	303
	refusal	0.0	94.4	5.6	306
	sametext	0.0	0.0	100.0	300
	think	0.6	52.4	46.9	309
	offtopic	0.0	0.7	99.3	300
flag	hint	0.0	28.4	71.6	303
	refusal	0.0	92.5	7.5	306
	sametext	0.0	0.0	100.0	300
	think	0.0	61.7	38.3	308
	offtopic	0.3	0.3	99.3	300
leaf	hint	0.3	28.4	71.3	303
	refusal	0.0	91.8	8.2	306
	sametext	0.0	0.0	100.0	300
	think	0.0	54.7	45.3	309
	offtopic	0.0	0.3	99.7	300
moon	hint	0.0	17.5	82.5	303
	refusal	0.0	90.2	9.8	306
	sametext	1.0	0.0	99.0	300
	think	0.0	47.9	52.1	309
	offtopic	0.0	0.0	100.0	300
wave	hint	0.0	24.1	75.9	303
	refusal	0.0	93.1	6.9	306
	sametext	0.0	0.0	100.0	300
	think	0.0	51.5	48.5	309
	offtopic	0.0	0.3	99.7	300

Table 13: Δ -LogitLens decodability for cooperative subjects with $\alpha = 0.5$.

Concept	Regime	Rank ws	Rank ns	$P(c^*)$	$\ \delta\ $	n
book	hint	469	3,457	9.7e-05	23.48	200
	refusal	76.9k	41.2k	3.5e-09	22.53	200
	sametext	119.3k	115.7k	4.7e-10	21.93	200
	think	25.5k	27.2k	1.6e-07	20.06	200
	offtopic	38.5k	53.6k	6.5e-08	17.07	200
flag	hint	25	19	2.30e-03	28.30	200
	refusal	2,503	321	9.1e-06	23.84	200
	sametext	41.5k	20.8k	5.3e-08	22.71	200
	think	755	126	8.7e-05	23.61	200
	offtopic	37.6k	79.5k	1.2e-07	17.03	200
leaf	hint	8	6	2.42e-03	24.23	200
	refusal	57	19	4.92e-04	23.30	200
	sametext	70	730	9.82e-04	21.61	200
	think	25	19	1.51e-03	20.75	200
	offtopic	34.7k	8,114	8.4e-08	17.35	200
moon	hint	4	27	1.96e-03	27.16	200
	refusal	22	183	4.53e-04	24.41	200
	sametext	3,181	4,013	1.3e-05	21.46	200
	think	11	237	4.32e-03	22.61	200
	offtopic	20.8k	66.1k	2.8e-07	17.46	200
wave	hint	5	22	0.016	26.03	200
	refusal	266	656	1.62e-04	22.40	200
	sametext	16.0k	44.0k	6.5e-07	20.89	200
	think	42	244	3.14e-03	21.24	200
	offtopic	38.9k	70.8k	5.8e-08	17.99	200

Table 14: Δ -LogitLens decodability for cooperative subjects with $\alpha = 1.0$.

Concept	Regime	Rank ws	Rank ns	$P(c^*)$	$\ \delta\ $	n
book	hint	965	4,046	2.0e-05	27.61	200
	refusal	26.7k	31.2k	3.9e-08	33.39	200
	sametext	97.7k	82.5k	2.3e-09	22.67	200
	think	7,476	12.0k	8.0e-07	33.71	200
	offtopic	47.1k	89.0k	6.5e-08	8.94	200
flag	hint	43	22	7.53e-04	29.69	200
	refusal	684	202	1.6e-05	32.53	200
	sametext	35.5k	21.1k	1.4e-07	20.98	200
	think	272	110	1.23e-04	34.15	200
	offtopic	3,770	3,789	1.6e-05	9.04	200
leaf	hint	6	3	9.09e-03	31.16	200
	refusal	13	21	3.21e-03	35.56	200
	sametext	138	461	3.55e-04	21.75	200
	think	4	9	0.021	34.91	200
	offtopic	734	1,103	8.2e-05	10.88	200
moon	hint	6	25	3.72e-03	31.78	200
	refusal	15	93	2.28e-03	35.38	200
	sametext	2,486	2,890	1.9e-05	21.26	200
	think	7	54	5.70e-03	35.87	200
	offtopic	352	1,135	3.42e-04	9.86	200
wave	hint	4	27	0.012	29.98	200
	refusal	22	294	3.44e-03	37.12	200
	sametext	1,815	12.1k	1.8e-05	20.26	200
	think	10	146	0.016	35.64	200
	offtopic	716	7,315	7.1e-05	10.52	200

Table 16: Δ -LogitLens decodability for strict subjects with $\alpha = 1.0$.

Concept	Regime	Rank ws	Rank ns	$P(c^*)$	$\ \delta\ $	n
book	hint	389	3,734	1.19e-04	25.18	200
	refusal	38.6k	113.6k	4.2e-08	30.04	200
	sametext	47.4k	87.3k	9.5e-08	16.77	200
	think	2,401	29.8k	2.2e-05	24.89	200
	offtopic	50.7k	86.2k	1.4e-10	18.84	200
flag	hint	20	27	6.23e-03	26.93	200
	refusal	89.7k	86.2k	3.8e-09	31.08	200
	sametext	4,602	3,675	1.2e-05	16.84	200
	think	3,508	10.3k	2.0e-05	26.81	200
	offtopic	5,687	44.6k	2.1e-08	20.18	200
leaf	hint	514	295	6.1e-05	26.95	200
	refusal	141.6k	132.8k	3.3e-11	30.13	200
	sametext	4,665	2,530	6.2e-06	18.17	200
	think	26.7k	21.9k	4.3e-07	26.33	200
	offtopic	88.8k	70.2k	1.1e-10	17.53	200
moon	hint	18	201	4.83e-03	27.72	200
	refusal	58.0k	96.5k	1.2e-08	28.93	200
	sametext	7,277	29.1k	5.9e-06	17.69	200
	think	41	1,023	1.11e-03	25.41	200
	offtopic	29.9k	115.0k	1.2e-09	18.83	200
wave	hint	2	13	0.048	25.86	200
	refusal	76.2k	52.0k	5.8e-09	30.34	200
	sametext	5,483	8,422	5.4e-06	16.94	200
	think	102	635	2.20e-04	25.84	200
	offtopic	64.2k	108.5k	2.4e-10	18.20	200

Table 15: Δ -LogitLens decodability for strict subjects with $\alpha = 0.5$.

Concept	Regime	Rank ws	Rank ns	$P(c^*)$	$\ \delta\ $	n
book	hint	668	4,166	5.5e-05	24.63	200
	refusal	48.8k	96.4k	1.6e-08	28.75	200
	sametext	69.3k	76.3k	2.5e-08	16.90	200
	think	4,274	26.8k	9.1e-06	23.72	200
	offtopic	41.4k	50.7k	8.0e-10	17.84	200
flag	hint	11	13	0.023	24.86	200
	refusal	73.2k	59.2k	5.3e-09	28.82	200
	sametext	949	334	5.3e-05	16.55	200
	think	482	1,482	1.91e-04	24.33	200
	offtopic	7,557	34.8k	3.8e-08	18.13	200
leaf	hint	1,615	2,393	2.3e-05	25.13	200
	refusal	144.3k	141.6k	1.7e-11	28.43	200
	sametext	24.8k	43.7k	4.5e-07	17.59	200
	think	29.2k	41.7k	3.0e-07	24.11	200
	offtopic	89.4k	67.2k	5.2e-11	17.72	200
moon	hint	39	389	8.46e-04	24.11	200
	refusal	47.4k	83.1k	1.8e-08	29.51	200
	sametext	2,587	7,854	3.1e-05	15.97	200
	think	402	4,958	1.57e-04	23.69	200
	offtopic	31.8k	109.4k	3.1e-09	16.96	200
wave	hint	6	15	0.033	24.44	200
	refusal	76.0k	36.8k	4.2e-09	29.07	200
	sametext	5,341	4,079	8.8e-06	16.84	200
	think	7,589	8,040	2.7e-06	24.92	200
	offtopic	57.7k	97.6k	4.9e-10	17.11	200

Table 17: Per-cell Base-AO recovery for cooperative subjects with $\alpha = 0.5$.

Concept	Regime	Exact %	Sem. %	$\overline{P(c^*)}$	Rank med.	n_j	n_p
book	hint	80	88	0.534	0	25	303
	refusal	36	48	0.229	35	25	304
	sametext	96	96	0.541	0	25	300
	think	36	36	0.249	54	25	308
	offtopic	0	0	3.51e-04	388	25	300
flag	hint	96	100	0.652	0	25	303
	refusal	56	56	0.328	3	25	305
	sametext	88	88	0.415	0	25	300
	think	56	56	0.352	3	25	309
	offtopic	0	0	1.47e-04	1,014	25	300
leaf	hint	76	84	0.475	0	25	303
	refusal	24	32	0.215	5	25	305
	sametext	72	76	0.267	0	25	300
	think	48	52	0.250	2	25	309
	offtopic	0	0	6.3e-05	2,493	25	300
moon	hint	100	100	0.585	0	25	303
	refusal	64	72	0.351	0	25	305
	sametext	76	76	0.355	0	25	300
	think	68	68	0.475	0	25	309
	offtopic	0	4	1.39e-03	120	25	300
wave	hint	84	84	0.501	0	25	303
	refusal	32	40	0.238	28	25	304
	sametext	60	60	0.130	0	25	300
	think	40	52	0.212	26	25	309
	offtopic	0	0	5.52e-04	321	25	300

Table 18: Per-cell Base-AO recovery for cooperative subjects with $\alpha = 1.0$.

Concept	Regime	Exact %	Sem. %	$\overline{P(c^*)}$	Rank med.	n_j	n_p
book	hint	92	96	0.747	0	25	303
	refusal	76	84	0.623	0	25	303
	sametext	96	96	0.587	0	25	300
	think	88	88	0.674	0	25	309
	offtopic	0	0	2.49e-04	511	25	300
flag	hint	100	100	0.758	0	25	303
	refusal	76	80	0.590	0	25	300
	sametext	52	52	0.286	0	25	300
	think	80	80	0.658	0	25	309
	offtopic	0	0	8.53e-04	1,163	25	300
leaf	hint	100	100	0.632	0	25	303
	refusal	64	64	0.482	0	25	303
	sametext	40	40	0.093	2	25	300
	think	76	76	0.515	0	25	309
	offtopic	0	0	1.81e-04	1,643	25	300
moon	hint	100	100	0.598	0	25	303
	refusal	72	76	0.499	0	25	304
	sametext	76	76	0.279	0	25	300
	think	84	88	0.581	0	25	309
	offtopic	0	0	1.10e-03	238	25	300
wave	hint	100	100	0.639	0	25	303
	refusal	80	88	0.535	0	25	305
	sametext	56	56	0.150	1	25	300
	think	80	80	0.527	0	25	309
	offtopic	0	0	1.37e-03	260	25	300

Table 20: Per-cell Base-AO recovery for strict subjects with $\alpha = 1.0$.

Concept	Regime	Exact %	Sem. %	$\overline{P(c^*)}$	Rank med.	n_j	n_p
book	hint	36	40	0.388	0	25	301
	refusal	0	0	7.40e-04	193	25	261
	sametext	0	0	0.018	10	25	300
	think	20	20	0.171	53	25	298
	offtopic	–	–	3.30e-04	400	–	300
flag	hint	44	48	0.387	0	25	298
	refusal	0	0	4.24e-04	273	25	237
	sametext	0	0	1.68e-04	689	25	300
	think	20	20	0.133	308	25	287
	offtopic	–	–	1.05e-04	1,165	–	300
leaf	hint	32	36	0.175	4	25	303
	refusal	0	0	1.96e-04	2,915	25	261
	sametext	0	4	7.32e-04	230	25	300
	think	16	16	0.099	566	25	301
	offtopic	–	–	6.4e-05	2,513	–	300
moon	hint	44	52	0.242	1	25	301
	refusal	0	0	2.61e-03	73	25	261
	sametext	0	0	4.62e-03	88	25	300
	think	24	24	0.164	13	25	290
	offtopic	–	–	6.97e-04	237	–	300
wave	hint	28	32	0.169	2	25	303
	refusal	0	0	3.27e-04	572	25	256
	sametext	0	0	2.65e-03	58	25	300
	think	20	24	0.053	140	25	296
	offtopic	–	–	5.42e-04	293	–	300

Table 19: Per-cell Base-AO recovery for strict subjects with $\alpha = 0.5$.

Concept	Regime	Exact %	Sem. %	$\overline{P(c^*)}$	Rank med.	n_j	n_p
book	hint	48	52	0.364	0	25	303
	refusal	0	0	1.23e-03	180	25	292
	sametext	8	8	0.032	9	25	300
	think	20	20	0.166	69	25	305
	offtopic	0	0	3.82e-04	342	25	300
flag	hint	40	44	0.411	0	25	302
	refusal	0	0	5.05e-04	274	25	291
	sametext	0	0	2.31e-04	619	25	300
	think	20	20	0.172	336	25	306
	offtopic	0	0	1.34e-04	1,114	25	300
leaf	hint	36	40	0.176	6	25	303
	refusal	0	0	5.14e-04	2,350	25	285
	sametext	0	8	4.67e-04	510	25	300
	think	20	24	0.097	608	25	308
	offtopic	0	0	6.3e-05	2,474	25	300
moon	hint	36	36	0.215	2	25	302
	refusal	4	8	8.59e-03	52	25	273
	sametext	8	20	9.19e-03	51	25	300
	think	24	24	0.121	15	25	306
	offtopic	0	4	9.72e-04	169	25	300
wave	hint	20	32	0.155	7	25	303
	refusal	0	0	2.32e-04	598	25	286
	sametext	0	12	1.16e-03	138	25	300
	think	12	28	0.047	476	25	305
	offtopic	0	4	5.81e-04	326	25	300

Table 21: Per-cell exact recovery (%) for cooperative subjects with $\alpha = 0.5$.

Concept	Regime	Base	Cross	Own
book	hint	72	68 [65,71]	63
	refusal	30	28 [27,29]	26
	sametext	96	34 [22,43]	30
	think	37	33 [30,36]	28
	offtopic	–	–	–
flag	hint	84	79 [78,79]	77
	refusal	43	39 [37,42]	37
	sametext	89	28 [16,39]	18
	think	47	42 [40,44]	41
	offtopic	–	–	–
leaf	hint	71	62 [60,64]	49
	refusal	30	27 [26,27]	22
	sametext	76	31 [21,44]	1
	think	43	30 [28,33]	17
	offtopic	–	–	–
moon	hint	92	90 [88,92]	84
	refusal	60	52 [48,58]	41
	sametext	94	61 [46,73]	26
	think	78	70 [64,77]	52
	offtopic	–	–	–
wave	hint	73	67 [66,68]	56
	refusal	37	36 [34,37]	26
	sametext	64	12 [6,20]	0
	think	35	30 [29,31]	23
	offtopic	–	–	–

Table 22: Per-cell exact recovery (%) for cooperative subjects with $\alpha = 1.0$.

Concept	Regime	Base	Cross	Own
book	hint	94	78 [65,92]	45
	refusal	77	59 [46,71]	21
	sametext	98	13 [0,35]	1
	think	85	59 [49,72]	27
	offtopic	0	0 [0,0]	0
flag	hint	96	83 [78,86]	47
	refusal	74	52 [49,55]	19
	sametext	63	1 [0,2]	0
	think	85	60 [53,67]	18
	offtopic	0	0 [0,0]	0
leaf	hint	90	49 [31,67]	14
	refusal	68	36 [19,49]	11
	sametext	34	0 [0,0]	0
	think	75	32 [19,48]	6
	offtopic	0	0 [0,0]	0
moon	hint	94	82 [77,89]	46
	refusal	76	57 [52,62]	32
	sametext	85	5 [1,10]	1
	think	87	69 [65,74]	32
	offtopic	0	0 [0,0]	0
wave	hint	95	61 [51,66]	22
	refusal	77	44 [39,48]	8
	sametext	60	0 [0,1]	0
	think	78	37 [27,41]	7
	offtopic	0	0 [0,0]	0

Table 24: Per-cell exact recovery (%) for strict subjects with $\alpha = 1.0$.

Concept	Regime	Base	Cross	Own
book	hint	55	49 [46,53]	49
	refusal	0	0 [0,0]	0
	sametext	9	7 [0,17]	1
	think	26	19 [16,24]	17
	offtopic	0	0 [0,0]	0
flag	hint	56	51 [50,52]	48
	refusal	0	0 [0,0]	0
	sametext	0	0 [0,0]	0
	think	17	15 [14,17]	14
	offtopic	0	0 [0,0]	0
leaf	hint	28	26 [23,30]	22
	refusal	0	0 [0,0]	0
	sametext	0	0 [0,0]	0
	think	14	13 [12,14]	12
	offtopic	0	0 [0,0]	0
moon	hint	46	47 [43,54]	38
	refusal	0	0 [0,0]	0
	sametext	0	1 [0,1]	0
	think	30	26 [24,29]	19
	offtopic	0	0 [0,0]	0
wave	hint	35	24 [20,27]	29
	refusal	0	0 [0,0]	0
	sametext	0	0 [0,0]	0
	think	11	6 [6,7]	7
	offtopic	0	0 [0,0]	0

Table 23: Per-cell exact recovery (%) for strict subjects with $\alpha = 0.5$.

Concept	Regime	Base	Cross	Own
book	hint	54	50 [48,51] ₂	46
	refusal	0	0 [0,0] ₂	0
	sametext	13	1 [0,2] ₂	1
	think	22	21 [20,22] ₂	20
	offtopic	—	—	—
flag	hint	56	55 [54,55] ₂	53
	refusal	0	0 [0,0] ₂	0
	sametext	0	0 [0,0] ₂	0
	think	22	21 [20,21] ₂	21
	offtopic	—	—	—
leaf	hint	26	28 [27,29] ₃	—
	refusal	0	0 [0,0] ₃	—
	sametext	0	0 [0,0] ₃	—
	think	13	13 [12,14] ₃	—
	offtopic	—	—	—
moon	hint	42	40 [39,41] ₃	—
	refusal	3	0 [0,0] ₃	—
	sametext	4	1 [0,2] ₃	—
	think	23	19 [17,19] ₃	—
	offtopic	—	—	—
wave	hint	33	35 [34,35] ₂	34
	refusal	0	0 [0,0] ₂	0
	sametext	0	0 [0,0] ₂	0
	think	12	9 [8,10] ₂	7
	offtopic	—	—	—

Table 25: Per-cell $\overline{P(c^*)}$ for cooperative subjects with $\alpha = 0.5$.

Concept	Regime	Base	Cross	Own
book	hint	0.534	0.564 [0.535,0.603]	0.498
	refusal	0.229	0.255 [0.238,0.273]	0.214
	sametext	0.541	0.226 [0.124,0.307]	0.174
	think	0.249	0.270 [0.248,0.299]	0.224
	offtopic	—	—	—
flag	hint	0.652	0.703 [0.691,0.715]	0.715
	refusal	0.328	0.351 [0.339,0.357]	0.346
	sametext	0.415	0.166 [0.092,0.206]	0.151
	think	0.352	0.360 [0.341,0.370]	0.370
	offtopic	—	—	—
leaf	hint	0.475	0.503 [0.480,0.532]	0.415
	refusal	0.215	0.227 [0.211,0.249]	0.187
	sametext	0.267	0.159 [0.075,0.222]	0.010
	think	0.250	0.244 [0.221,0.264]	0.147
	offtopic	—	—	—
moon	hint	0.585	0.614 [0.571,0.679]	0.632
	refusal	0.351	0.338 [0.322,0.370]	0.301
	sametext	0.355	0.344 [0.275,0.405]	0.133
	think	0.475	0.459 [0.421,0.506]	0.394
	offtopic	—	—	—
wave	hint	0.501	0.526 [0.474,0.562]	0.451
	refusal	0.238	0.277 [0.247,0.301]	0.213
	sametext	0.130	0.030 [0.018,0.052]	7.97e-04
	think	0.212	0.236 [0.220,0.248]	0.178
	offtopic	—	—	—